

International Institute of Information Technology, Bangalore
CSE 511 Algorithms: Practice Problem

You are given an array $A[1 \dots N]$ consisting of N integers. You must process Q operations of the following three types.

1. Range Increment

An operation of the form

$$1 \ l \ r \ v$$

adds v to every element whose index lies in the range $[l, r]$. Formally,

$$A[i] \leftarrow A[i] + v \quad \text{for every } l \leq i \leq r.$$

2. First Threshold Position

An operation of the form

$$2 \ l \ r \ x$$

asks for the smallest index j satisfying

$$l \leq j \leq r \quad \text{and} \quad A[j] \geq x.$$

If no such index exists, output -1 .

3. Range Maximum

An operation of the form

$$3 \ l \ r$$

asks for the maximum value in the range $[l, r]$. That is, output

$$\max_{l \leq i \leq r} A[i].$$

Input Format

The first line contains two integers N and Q , representing the size of the array and the number of operations, respectively.

The second line contains N integers

$$A[1], A[2], \dots, A[N],$$

representing the initial values of the array.

Each of the next Q lines describes one operation in one of the following formats:

- **1 1 r v**: add v to every element in the range $[l, r]$.
- **2 1 r x**: find the smallest index $j \in [l, r]$ such that $A[j] \geq x$.
- **3 1 r**: find the maximum value in the range $[l, r]$.

Output Format

For every operation of type 2, print the smallest valid index j . If no such index exists, print -1 .

For every operation of type 3, print the maximum value in the specified range.

Print the answer to each query on a separate line.

Constraints

$$1 \leq N, Q \leq 2 \times 10^5$$

$$-10^9 \leq A[i] \leq 10^9$$

For every operation of type 1,

$$1 \leq l \leq r \leq N, \quad -10^9 \leq v \leq 10^9.$$

For every operation of type 2,

$$1 \leq l \leq r \leq N, \quad -10^9 \leq x \leq 10^9.$$

For every operation of type 3,

$$1 \leq l \leq r \leq N.$$

Sample

Sample Input

```
7 8
2 1 5 3 4 1 6
2 2 7 5
1 2 5 3
2 2 7 7
3 1 5
1 1 7 -4
2 1 7 5
1 6 7 5
2 4 7 4
```

Sample Output

```
3
3
8
-1
7
```